For Examiner's Use

4 **(a)** "The magnitude of *gravitational field strength* at a point is proportional to the *gravitational potential gradient* at that point."

Explain the terms in italics.

gravitational field strength: .....

.....

gravitational potential gradient: .....

..... [2]

(b) Fig. 4.1 shows four equipotential lines at 1000-km intervals in a non-uniform gravitational field.

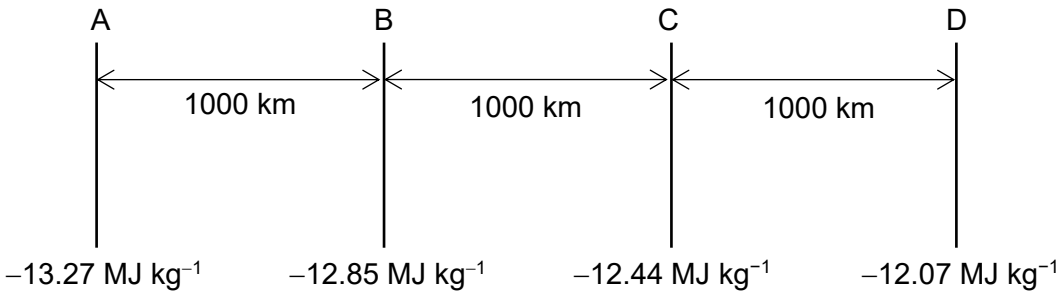

Fig. 4.1

(i) Draw an arrow on Fig. 4.1 to indicate the direction of the gravitational field. [1]

(ii) Calculate the ratio of

$$\frac{\text{field strength at B}}{\text{field strength at C}}.$$

ratio = ... [3]

For Examiner's Use

(c) A theoretical long open tube is placed in the gravitational field as shown in Fig. 4.2. An object is released from rest at B.

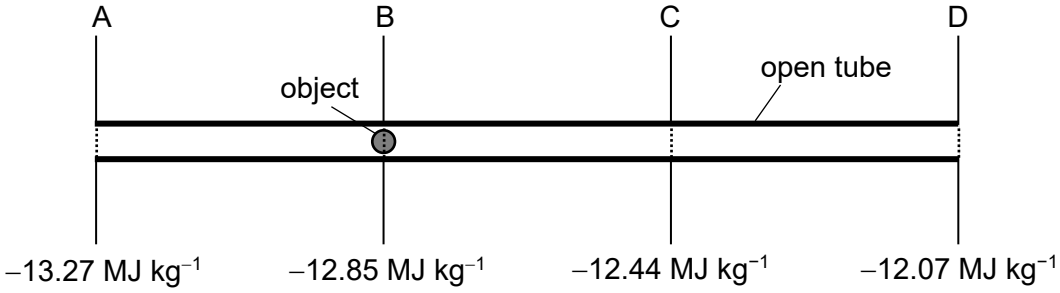

Fig. 4.2

Determine the speed with which the object leaves the tube.

speed = ... m s^{-1} [4]